01 · LESSON

Plant products: starch, lipids and proteins

9B Plant growth · Lesson 4 of 7

Date _____ Class _____

Today I can 今天我能

- ☐ I can describe how plants use glucose to make starch, cellulose, lipids and proteins, and explain why nitrates are needed.
- ☐ I can test a leaf for starch and explain what happens when a seed germinates.

What happens today

1. Do Now: potato and iodine
2. Learn: Carbohydrates: starch and cellulose · Lipids and proteins · and more
3. Practical: Testing a variegated leaf for starch
4. Challenge, then the exit ticket

Student Book pages 26 to 27 · Workbook pages 19

What I learned today (one or two sentences) 今天我学到了：

02 · KEY VOCABULARY

Key Vocabulary

Term	中文	Definition
starch	淀粉	a polymer made of many glucose molecules joined together; a plant's energy store
polymer	聚合物	a long molecule made of many small molecules joined in a chain
cellulose	纤维素	a polymer of glucose used to make plant cell walls
lipid	脂质	a fat or oil; insoluble in water
protein	蛋白质	a polymer made of amino acids; enzymes are proteins
amino acid	氨基酸	a small molecule that proteins are built from; needs nitrogen
enzyme	酶	a protein that speeds up a reaction in a living thing
nitrate	硝酸盐	a mineral salt that supplies nitrogen for making amino acids
germination	发芽	when a seed starts to grow into a seedling
iodine solution	碘液	an orange-brown liquid that turns blue-black with starch
Benedict's solution	本尼迪克特试剂	a blue liquid that turns orange-red when heated with glucose

Command words 指令词: Complete 完成（填写） · Draw 画出 · Explain 解释 · Label 标注 · Match 连线（配对） · Name 说出名称 · Predict 预测 · Use 使用（运用）

03 · QUICK REVIEW

Quick Review

potato and iodine

1 What colour change would you see if you added iodine solution to a slice of potato?

2 Which substance in the potato causes this change?

3 Which substance made in photosynthesis was used to make it?

04 · CORE

Core 1 of 4

Everyone does these. Answer on the card or under it. 人人都做，答案写在卡片上或下面。

1 Practical: testing a variegated leaf for starch.

a) Draw the leaf before the test and after adding iodine. Label the green and white parts on the first drawing, and the blue-black parts on the second.

Before the test	After adding iodine

Core 2 of 4

Everyone does these. Answer on the card or under it. 人人都做，答案写在卡片上或下面。

1 *continued*

b) Name the solution used to test for starch and the colour it turns when starch is present.

c) Explain why only some parts of the leaf changed colour.

Core 3 of 4

Everyone does these. Answer on the card or under it. 人人都做，答案写在卡片上或下面。

2 Complete the sentences. Use the word bank.

Glucose molecules made in photosynthesis are joined together to make a polymer called _____.

Starch is an energy store. It is found in the chloroplasts, in storage organs such as _____, and in seeds.

Sugars are also used to make _____, which builds plant cell walls, and _____ such as the oils in seeds.

Core 4 of 4

Everyone does these. Answer on the card or under it. 人人都做，答案写在卡片上或下面。

3 Match each substance to one of its uses in a plant. Draw a line.

starch	speed up reactions such as photosynthesis and respiration
cellulose	energy store in seeds and potatoes
lipids	build cell walls
proteins	built from amino acids using nitrogen from nitrates
enzymes	make the waterproof cuticle and cell membranes

Challenge 1 of 2

When Core is finished. Full sentences. 完成基础题后做；请写完整句子。

C1 A plant was left in the dark for 24 hours. A metal foil stencil was then attached to one leaf and the plant was left in bright light for a day. The leaf was then tested for starch.

a) Explain why the plant was left in the dark first.

b) Predict the result of the starch test and explain it.

Challenge 2 of 2

When Core is finished. Full sentences. 完成基础题后做；请写完整句子。

C1 *continued*

c) Explain why a metal foil stencil was used and not a clear plastic one.

C2 Avocado fruit flesh is rich in lipids. Avocado seeds also contain lipids. Explain why the plant puts lipids in each place. The two reasons are different.

Exit Ticket

Hand this card in at the end of the lesson.

Name _____

1 Which solution tests for starch, and what colour does it turn?

2 Why do seeds need a store of starch?

3 What are enzymes made of, and which mineral salt do plants need to make them?

Today I can: Goal 1 ☐ Goal 2 ☐